

## a From atomic composition to 3D coordinates

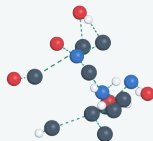
INPUT

$\text{C}_{10}\text{H}_9\text{N}_3\text{O}_5$

Atomic identities  
Charge 0 · Singlet

No bond graph

Harmonic source



$X_0$

Flow time 0.34



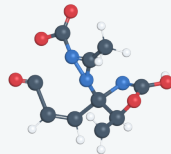
$X_t$

Flow time 0.69



$X_t$

Generated molecule



$X_{\text{gen}} \in \mathbb{R}^{27 \times 3}$

At each step: coordinates + time + composition  $\rightarrow$  atom-wise velocity

## b Learning physical corrections during training

Generated training anchors

Energy-model forces

Nearby force-shifted targets

$\theta_{\text{physical}}$

Original anchor targets

$\theta_{\text{replay}}$

ONE FINAL NETWORK

$\theta_{\text{base}} + \theta_{\text{physical}} - \theta_{\text{replay}}$

128 forward passes · No oracle at inference